Course code :- BMAT205L

Course Name :- Discrete Mathematics And Graph Theory.

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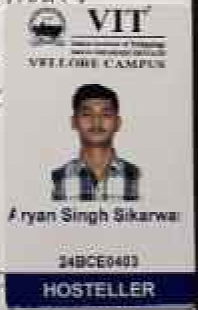
Tutorial - 4
(module - 2)

Que-1 Let R be the Additive group of real numbers and let $\mathbb{R}^+$ be the multiplicative group of positive and real numbers.
Define :- $f \to \mathbb{R} \to \mathbb{R}^+$ by $f(x) =$
for all $x \in R$. prove that $f$ is
Homomorphism.



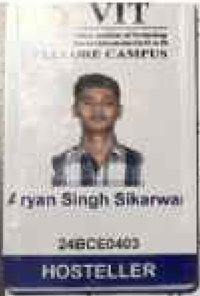

Solve → The function $f : \mathbb{R} \to \mathbb{R}^+$

$f \to (G_1 \to G_2)$ b/w two group

a group Homomorphism. $a, b \in G$.

$$f(a+b) = f(a) \cdot f(b)$$

where $(+)$ is the operation in $G_1$ and $(\cdot)$ is the operation in $G_2$.




Identifying the groups and their operations :- (2)

$(\mathbb{R},+)$ : operation in Addition.

$= (\mathbb{R}^+, \times)$ : operation in Multiplication.

$+y) = f(x) \times f(y)$

for all $x, y \in \mathbb{R}$.

Step-3 Substitute the given function :-

$$f(x+y) = e^{x+y}$$

$$f(x) \cdot f(y) = (e^x) \cdot (e^y) = e^{x+y}$$

→ on comparison :-

$$f(x+y) = e^{x+y} = e^x \cdot e^y = f(x) \cdot f(y)$$

the Homomorphic property is satisfy $(x, y \in \mathbb{R})$

(1) $(\mathbb{R},+) : 0$ :- $\quad f(0) = e^0 = 1$

(2) $(\mathbb{R},\times)$ is $-x$ :- $\quad f(-x) = e^{-x} = \frac{1}{e^x} = [f(x)]^{-1}$

showing inverse also correspondence property.

Therefore, $f(x) = e^x$ is a group Homomorphism from $(\mathbb{R},+)$ to $(\mathbb{R}^+, \times)$ Ans


Que-2 - e Additive group of real 3
numbe group T, be the
multip p of complex numbers
with ue 1. Define $f : R \to T$
by + $i\sin(\theta)$. prove that
$f$ i Homomorphism.

Solve Show $f : R \to T$ given by
$f(\theta) = \sin\theta + i\sin\theta$ is a
group Homomorphism.

Here $T = \{z \in C : |z| = 1\}$ with
multiplication, domain $(R, +)$.

Using Euler property :-

→ $\cos\theta + i\sin\theta = e^{i\theta}$, Then for
$s, \theta \in R$

$$f(s+\theta) = e^{i(s+\theta)} = e^{is} \cdot e^{i\theta} = f(s)f(\theta)$$

Therefore $f$ preserves the operation.
→ $f$ is a group Homomorphism.

Identity: $f(0) = 1$. Inverse
$f(-\theta) = e^{-i\theta} = (e^{i\theta})^{-1}$

Que-3 is a Homorphism of a (4)
Quoc H is a subgroup of G,
then $f(H)$ is a subgroup


Solve :- G' be a Homomorphism

prove $f(H)$ is a subgroup of $G'$.

Proof :-

① nonempty :- $e_G \in H$ because H is a Subgroup. Then $f(e_G) = e_{G'} \in f(H)$

So $f(H)$ is nonempty.

② closure under product :- take $a, b \in f(H)$

Then $a = f(h_1)$, $b = f(h_2)$ with $h_1, h_2 \in H$. Since f is a Homomorphism.

$$ab = f(h_1) f(h_2) = f(h_1 h_2)$$

But $h_1 h_2 \in H$ (subgroup), so $ab \in f(H)$

(2) Closur... nverse :-


Let $a \in$ ... $f(h)$ for some $h \in$ ... $a^{-1} = (f(h))^{-1} = f(h^{-1})$.

Since ... $^{-1} \in f(H)$

All S... ditions hold ... $G'$. Therefore,

$f(H)$ is a subgroup of $G'$.

<u>Que-4</u> :- find the codeword generated by a parity check matrix.

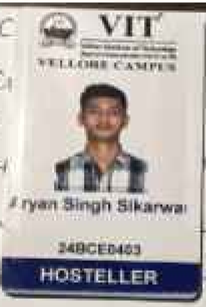
$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

when the encoding function $e : B^3 \to B6$

<u>Sol^n</u> The parity check Matrix can be written as :-

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

So expect $2^n = 2^3 = 8$ codewords.

Let C  (C4, C5, C6) ⑥



① $C_1$

② $C_2$

③ $C_1 +$ ... $0$

from ... $+ C_4$

from ... $+ C_4$

$$C_2+C_4 = C_3+C_5 \Rightarrow C_2+C_4+C_5$$

from ③ $C_1+C_2+C_3+C_6=0$

Substitute $C_1 = C_2+C_4$ and $C_3$

$$(C_2+C_4)+C_2+(C_2+C_4+C_5)+C_6=0$$

$$C_2+C_5+C_6=0 \Rightarrow C_2 = C_5+C_6$$

<u>Now Substituting Back</u> :-

$C_2 = C_5+C_6$

$C_1 = C_2+C_4 = C_4+C_5+C_6$

$C_3 = C_4+C_6$

$C_4 = C_4$ , $C_5 = C_5$ and $C_6 = C_6$

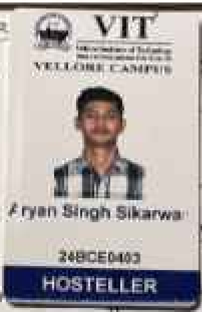
→ <u>All final Keywords Are</u> :-

000000, 001011, 01110, 010101, 101100

100111, 110010, 111001 Ans

Que-5 for a C(6,3 ... generated (7)
matrix G is ... ow :-

$$\begin{bmatrix} 1 & 0 & \\ 0 & 1 & \\ 0 & 0 & \end{bmatrix}$$




(i) construct the ... ck matrix :-

Solve → for a Sy... or $G = [I_k | P]$

$H = [P^T | I_{n-k}]$

compute $P^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$ → parity check matrix.

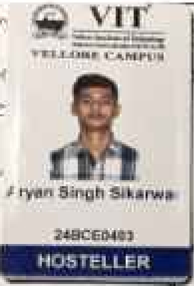
(ii) find all codewords Generated by G.

→ All codewords are $2^k = 2^3 = 8$ key codewords

$G \cdot e(000) = 000000$
$G \cdot e(001) = 001110$
$G \cdot e(010) = 010011$
$G \cdot e(011) = 011101$
$G \cdot e(100) = 100101$
$G \cdot e(101) = 101011$

$G \cdot e(110) = 110110$
$G \cdot e(111) = 111000$

The codewords are
000000, 001110, 0100..
011101, 100101, 1010..
110110, 111000 Ans

(iii) Minin 


g Distance :- ⑧

→ Comput istance weight of each non-z d.

(iv) Erro le :-

→ A codeword with distance $d_{min}$ detects upto $d_{min}-1$ errors

$$3-1 = 2 \text{ errors}$$

(v) Errors correctable :-

→ A code corrects upto $\left[\frac{d_{min}-1}{2}\right]$ errors.

$$\frac{3-1}{2} = \frac{2}{2} = (1) \rightarrow \text{error can be corrected.}$$

(vi) Received word r = 100011 . find the error And correct.

2) $s = Hr^T \bmod 2 \rightarrow r = (1,0,0,0,1,1)^T$

$$s = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$r \Rightarrow 100011$ Now corrected $r = 101011$

Ans

Que-6 Find ... ds Generated (9)
by the € ... tion
$e: B^3 \to$ ... spect to the
parity c ... .


$$\begin{bmatrix} \ldots \\ 0\ 1\ 0 \\ 0\ 0\ 1 \end{bmatrix}$$

Solve
$$\Rightarrow \quad H = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Let $C = (C_1 \ldots\ldots C_6)$

Total Codewords
$\Rightarrow 2^n = 2^3 = 8$ codewords

① $C_3 = 0$
② $C_2 + C_1 + C_4 = 0$
③ $C_1 + C_2 + C_3 + C_6 = 0$

$C_4 = C_5 + C_6$
$C_1 = C_2 + C_4 = C_2 + C_5 + C_6$
$C_3 = 0$

The final total 8 codewords are :-

{000000, 000011, 010101, 010110, 100101,
100110, 110000, 110011} Ans

fall - 2025 - 2026 (10)

BMAT205L
Discrete - Mathematics
Tutorial - 6 (module - 3)

Name :- Aryan Singh Sikarwar
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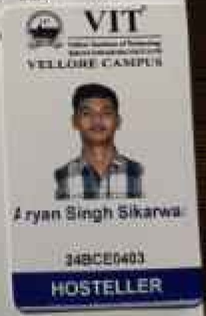
Que-1 Suppose that a person deposits \$10,000 in a savings account at a bank yielding 11 % per year with interest compounded annually, How much will be in the account after 30 year




Solve- $A = P \cdot (1 + r)^t$

$P = \$10000$

$r = 11\% = 0.11$

$t = 30$ years

$A = 10000(1+0.11)^{30}$

$A = 10000(1.11)^{30}$

$\rightarrow (1.11)^{30}$ and then multiply by 10000

$A = 10000(22.89229 6)$

$A = 228922.96$ after 30 years

Ans

Que-2 A comp 

...nsidered a string (11)
of decimal ... valid codeword if
it contain ... number of 0 digits.
for instan... 869 is valid,
whereas ... is not valid.
Let $a_n$ ... er of valid n-digit
codeword ... currence relation
for $a_n$.

Solve- Each Digit can be 0-9 (10 possible)

$$a_{n+1} = 9a_n + b_n$$

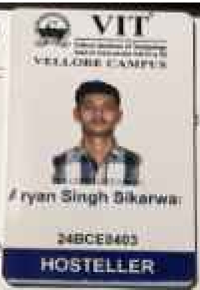
$$b_{n+1} = 9b_n + a_n$$

Since, $a_n + b_n = 10^n$, we can eliminate $b_n$:

$$a_{n+1} = 9a_n + (10^n - a_n) = 8a_n + 10^n$$

Recurrence :- $a_{n+1} = 8a_n + 10^n$, $a_1 = 9$ Ans

Que-3 Suppose that the number of bacteria in a colony triples every hour. (a) set up a recurrence relation for bacteria after (n) hours have elapsed. (b) If 100 bacteria are used to begin a new colony, how many bacteria will be in the colony in 10 Hours?




Solve - Bacteria hour :-

Let $N_t$ = t hours

(a) Each ule :-

(b) $N_0 = 10$

Then, after 10 hours :-

$$N_{10} = 100 \times 3^{10} = 100 \times 59049$$

$$N_{10} = 5904900$$

$\Rightarrow 5.9 \times 10^6$ Bacteria Ans

Que-4 How many solutions does the equation $x_1 + x_2 + x_3 + x_4 = 17$, where $x_1, x_2, x_3$ and $x_4$ are non-negative numbers?

Solve - Equation :- $x_1 + x_2 + x_3 = 11$

Number of non-negative integer shown solutions are :-

$$\binom{11+3-1}{3-1} = \binom{13}{2} = 78$$

$\Rightarrow$ 78 solutions Ans

Que-5 How m 

are there to the Equa ... $x_3 + x_4 = 17$, (13)

where $x_1$ ... $x_4$ are non-negati...

Solve -: $x_1$ ... $x_4 = 17$

Count $= \binom{\ldots}{4-1} = \binom{20}{3} = 1140$

$\Rightarrow$ 1140 solutions Ans

Que-6 How many solutions are there to the Equations $x_1 + x_2 + x_3 + x_4 + x_5 = 21$, where $x_i$, $i = 1, 2, 3, 4, 5$ is a non-negative integer such that (a) $x_1 > 1$
(b) $x_i \geq 2$ for $i = 1, 2, 3, 4, 5$

Solve -:
$y_1 = x_1 - 2$, $y_2 = x_2 - 1$
$y_3 = x_3 - 4$, $y_4 = x_4$, $y_5 = x_5$

All $y_i \geq 0$

Then $(y_1 + 2) + (y_2 + 1) + (y_3 + 4) + y_4 + y_5 = 21$

$y_1 + y_2 + y_3 + y_4 + y_5 = 14$

$\binom{14+5-1}{5-1} = \binom{18}{4} = 3060$

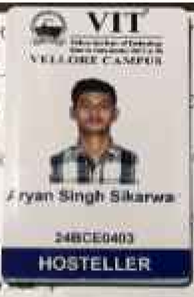
$\Rightarrow$ 3060 solutions Ans

Que-7 How m ngs of length (14)
8 either sta 1 or end
with th 0?

Solve-:- Let of valid binary
of length n with
ive 0s.

So Recurre

$$a_n = a_{n-1} + a_{n-2}$$




with base case,

$a_1 = 2$("0", "1")

$a_2 = 3$("01", "10", "11")

that the fibonacci series sequence method:-

$a_3 = 5$, $a_4 = 8$, $a_5 = 13$, $a_6 = 21$, $a_7 = 34$

$a_8 = 55$

Therefore, 55 Binary strings of length 8 with no consecutive 0s.

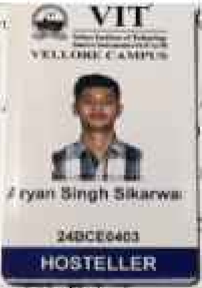
Que-8 There are 345 students at a college who have taken a course in calculus, 212 who have taken a course in discrete and 186 who have taken course in both calculus and discrete maths,

How many st                  taken a course (5)
in either                  Discrete maths?

Solve -: |C ∪                  A|+|D| - |C∩A| - |C∩D|
                  | + |C∩A∩D|

                  +260 - 120 - 90 - 80 + 50

                  + 50




∴ 575, students took at least one course Ans

Que - 9 find the number of positive number not exceeding 100 that are not are divisible by 5 or 7?

Solve :- number of integers Divisible by 5 $\Rightarrow \left[\frac{100}{5}\right] = 20$

number Divisible by 7

$\Rightarrow \left[\frac{100}{7}\right] = 14$

Both by 5 or by 7 is 35

$\left[\frac{100}{35}\right] = 2$

Total = Div(5) + Div(7) - Div(5 and 7)

= 20 + 14 - 2

= 32 Ans

Que-12 Let S = {2, 


is ordered by divisibility ... Hasse Diagram ... the maximal and ... element of S. 

Sol^n :- Let S = { ... } ordered ... ty.

Covering ... (2,6), (4,12), (4,20), (6,12)


[Hasse Diagram]

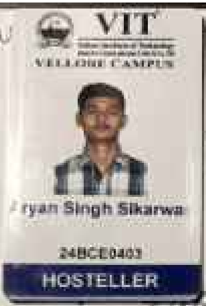
→ minimal = {2}

→ maximal = {12, 20} Ans

Que-14 How many different strings can be made from the letters MISSISSIPPI using all the letters?

Sol^n :- Letters are multiple in (MISSISSIPPI)

M=1, I=4, S=4, P=2 {Total = 11}




N ...tinct permutations of ...itset :-

$\frac{11!}{1!4!2!4!}$

$\frac{11!}{1!\times 4!\times 4!\times 2!} = 34,650$

∴ 34,650 Distinct strings ans

Que-15 How many different strings can be made from the letter ORONO, using some or all of the letters?

Solve :- (ORONO) ⇒ O = 3, R=1, N=1

By formula :- $\frac{(a+r+n)!}{a!r!n!}$

Length 1 : 3 strings
Length 2 : 7 strings
Length 3 : 13 strings
Length 4 : 20 strings
Length 5 : 20 strings

Sum ⇒
3+7+13+20+20 = 63

∴ Total 63 Distinct strings (using some or all letters)
ans

Fall - 2025 - 2026

BMAT205L

Discrete - Mathematics

- tutorial - 5 (module - 3)

Name :- Aryan Singh Sikarwar

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Que-1 In a class of 50 students,
play football, and 16 students
hockey. It is found that 10
play Both games. find the numb
students who play neither.

Solve :- football = 20
Hockey = 16
football and Hockey = 10
Both

- Number who play atleast one

$\Rightarrow 20 + 16 - 10 = 26$

Now, $\Rightarrow$ Total - (at least one)
(neither) $\Rightarrow 50 - 26$
$\Rightarrow$ 24 Students Ans

Que-2 How m ... there to select five ... om a 10-member ... o make a trip to a ... nother school? (19)


Solve - Choos ... 10 $\Rightarrow \binom{10}{5}$

$\binom{10}{5}$ ... 252

$\therefore$ 252 ways Ans

Que-3 Find the number of bit strings of length 10 that either begin with 1 or end with 0.

Solve :- Strings that fix with 1 : remain $9 = 2^9$

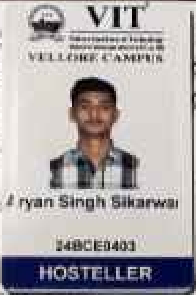
Strings that end with 0 : remain $9 = 2^9$

Strings that end both begin with 1 AND end with 0 : fix both $\rightarrow$ remain 8 bits $= 2^8$

$\Rightarrow 2^9 + 2^9 - 2^8 = 512 + 512 - 256$

$= 768$

$\therefore$ 768 bit-strings Ans

Que-4 The 

auditorium
are to be ... ith an
Uppercase ... etter followed
by a po... not exceeding 100.
What is ... number of chairs
that can ... differently?

Solve :- There ... upper cases letters
and integers 1 through 100 (100 choices)
Every Distinct (letter, number) pair
is a Distinct label.

Total = 26 × 100
= 2600

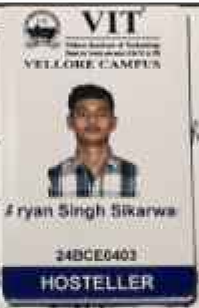
∴ 2600 Different labels Ans

Que-6 If 200 faculty members are
speak french and 50 speak
Russian and 20 can speak both
french and Russian. find How
many faculty members can speak
either french or Russian?

Solve -: Union = 200 -
Union = 240

formula ⇒ french - ... french or Russian

∴ 240 faculty -




Que-7 How many p... n be made out of the letters of the word 'BASIC'? How many of these

(i) begin with B?
(ii) end with C?
(iii) (B) and (C) occupy the end place?

Solve :- (i) Total permutation $= 5! = 120$

Begin with B : fix first $\rightarrow 4! = 24$

(ii) End with C : fix last $\rightarrow 4! = 24$

(iii) B and C occupy the ends : two choices which gets B vs C, and permute the remaining 3 letters in the middle

$2 \cdot 3! = 2 \cdot 6 = 12$

⇒ (i) 24 (ii) 24 (iii) 12 Ans

Que-10 How [illegible] tive integers not exc[illegible] are divisible by non[illegible] and 11? (22)


Solve -: Co[illegible] by at least one [illegible] from 1000.

① $\left[\frac{1000}{3}\right]$ [illegible] pairs (lcm): 21 → 47, 33 → 30, 77 → 12

② $\left[\frac{1000}{7}\right] = 142$

Sum pairs = 47 + 30 + 12 = 89

③ $\left[\frac{1000}{11}\right] = 90$

Triple lcm $3 \times 7 \times 11 = 231$

$231 \to \left[\frac{1000}{231}\right] = 4$

So Divisible by $\geq 1 \Rightarrow 333 + 142 + 90 - 89 + 4$

$\Rightarrow 480$

Thus the divisible by none $\Rightarrow 1000 - 480$

$\Rightarrow 520$

$\therefore$ 520 positive integers Ans

Que-11 Use generating functions to solve the following recurrence relations :-

(i) $a_n - 2a_{n-1} - \ldots$, $n \geq 2$ with $a_0 = 3, a_1 = 1$


Solve :- Charact... equation :-

$\Rightarrow r^2 - 2r \ldots$

$\Rightarrow (r-3)(r+1) = 0$

$\Rightarrow r = 3, -1$

General $A_n = A \cdot 3^n + B(-1)^n$. using initial conditions

$n=0 : A+B = 1$

$n=1 : 3A - B = 3$

Solving : $\Rightarrow A = 1, B = 0$. Hence

$\{A_n = 3^n\}$ - Ans

(ii) $y_{n+2} - 4y_{n+1} + 3y_n = 0$, $y_0 = 2$, $y_1 = 4$

Solve :- Characteristic Equation :-

$\Rightarrow r^2 - 4r + 3 = (r-1)(r-3)$

General $A_n = A + B \cdot 3^n$ : from initials,

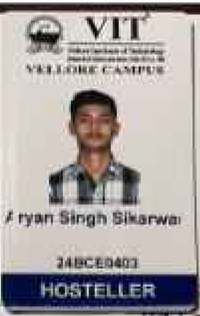
$\Rightarrow A+B=2$

$A+3B=4$

Now After solving

$A=1\ ,\ B=1$

$\Rightarrow \{An = 1+3^n\}$ Ans




(P) $a_n = 8a_{n-1} - 16a_{n-}$ $=16$, $=80$

Solve- Characteristic Equation :-

$\Rightarrow r^2 - 8r + 16$

$\Rightarrow (r-4)^2 \rightarrow$ Double root 4

General :- $a_n = (A+Bn)4^n$

$\Rightarrow$ Use :-

$n=0$ : $A=16$

$n=1$ : $(A+B)4 = 80$

$\therefore A+B=20$

$(A=16)$ and $(B=4)$

$\{An = (16+4n)4^n\}$ Ans

(viii) $a_n = 3a_{n-1} + 1$, $n \geq 1$ $(a_0 = 1)$ 

Solve :- $a_n = 3a_{n-1} + 1$, $(a_0 = 1)$

Homogeneous $(C \cdot 3^n)$

constant particular (C) satisfies

$$C = 3C + 1$$

$$C = -\frac{1}{2}$$

$$\left\{ a_n = A \cdot 3^n - \frac{1}{2} \right\}$$


→ using initial conditions :-

use $a_0 = A - \frac{1}{2} = 1 \Rightarrow A = \frac{3}{2}$

$$\left\{ A = \frac{3}{2} \right\}$$

$$\left\{ a_n = \frac{3}{2} \cdot 3^n - \frac{1}{2} = \frac{3^{n+1} - 1}{2} \right\} \text{ Ans}$$

(ix) $a_{n+2} - 2a_{n+1} + a_n = 2^n$, $(n \geq 0)$

$$\{ a_0 = 2, \ a_1 = 1 \}$$

Solve - Homogeneous part has double root

1 → general homogeneous

$$\{ A + Bn \}$$

$$\{A_n = A + Bn + 2^n\}$$

- Use ($n=0$)

$A + 1 = 2$

$\{A = 1\}$ use $n=1$: $1 + B + 2 = 1$

$B = -2$

$$\{A_n = 1 - 2n + 2^n\}$$ Ans